Problem A. Ascending Matrix

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	1024 mebibytes

You are given integers N, M, K, R, C, and V. Find the number of N by M integer matrices $a = (a_{i,j})$ that satisfy all of the following conditions, modulo 998244353.

- $1 \le a_{i,j} \le K$ for all $1 \le i \le N$, $1 \le j \le M$.
- $a_{i,j} \le a_{i,j+1}$ for all $1 \le i \le N$, $1 \le j \le M - 1$.
- $a_{i,j} \le a_{i+1,j}$ for all $1 \le i \le N - 1$, $1 \le j \le M$.
- $a_{R,C} = V$.

Input

The first line contains integers N, M ($1 \le N, M \le 200$), K ($1 \le K \le 100$), R ($1 \le R \le N$), C ($1 \le C \le M$), and V ($1 \le V \le K$).

Output

Print the answer.

Examples

standard input	standard output
2 2 2 1 1 1	5
2 2 2 1 2 1	3
4 5 6 2 3 4	3700125
200 100 100 70 60 30	546626227

Problem B. Bit Operation

Input file:	*standard input*
Output file:	*standard output*
Time limit:	2 seconds
Memory limit:	1024 mebibytes

You are given an integer array A of length N, consisting of 0's and 1's. Let a be initially the array A. You are going to perform the following operation $N-1$ times.

- Let n be the current length of a. Choose an integer i $(1 \le i \le n-1)$ and delete the i-th and the $(i+1)$-th elements of a. Then, by letting x and y be the deleted elements, insert either $x \,\&\, y$ or $x \mid y$ to the position of the deleted elements. Here $x \,\&\, y$ and $x \mid y$ denote the bit-AND and bit-OR operations, respectively.

There are $2^{N-1} \times (N-1)!$ ways to perform the operations. Count the number of ways that result in a single value of 1, modulo 998244353.

Input

The first line contains an integer N $(1 \le N \le 10^6)$.

The second line contains integers $A_1, A_2, \ldots, A_N$ $(0 \le A_i \le 1)$.

Output

Print the answer.

Examples

standard input	standard output
3 0 1 0	2
5 1 1 1 1 1	384
7 0 1 1 0 1 0 1	25515

Problem C. Count Min Ratio

Input file:	*standard input*
Output file:	*standard output*
Time limit:	4 seconds
Memory limit:	1024 mebibytes

You have R red balls, B blue balls, and one green ball. You are going to arrange the balls in a row. The score of an arrangement is defined as follows:

- Let $l_{\mathrm{R}}, l_{\mathrm{B}}, r_{\mathrm{R}}, r_{\mathrm{B}}$ be the number of red/blue balls to the left/right of the green ball, respectively. Then, the score is the maximum integer x such that $l_{\mathrm{B}} \times x \le l_{\mathrm{R}}$ and $r_{\mathrm{B}} \times x \le r_{\mathrm{R}}$.

Find the sum of scores of all possible arrangements, modulo 998244353. Note that balls of the same color cannot be distinguished, thus two arrangements are considered different if and only if there is such an i that the color of the i-th ball in the first arrangement differs from that of the second.

Input

The first line contains integers R ($1 \le R \le 10^{18}$) and B ($1 \le B \le 10^{6}$).

Output

Print the answer.

Examples

standard input	standard output
10 3	8390
3 10	0
100 10	801171977
999999999999999999 999999	448294209

Problem D. Do Use FFT

Input file:	*standard input*
Output file:	*standard output*
Time limit:	10 seconds
Memory limit:	1024 mebibytes

You are given integer sequences A, B, and C, each of length N. For each $k = 1, 2, \ldots, N$, find the following value modulo 998244353.

$$\sum_{1 \le i \le N} \left(C_i \times \prod_{1 \le j \le k} (A_i + B_j) \right)$$

Input

The first line contains an integer N ($1 \le N \le 250000$).

The second line contains N integers $A_1, A_2, \ldots, A_N$ ($0 \le A_i < 998244353$).

The third line contains N integers $B_1, B_2, \ldots, B_N$ ($0 \le B_i < 998244353$).

The fourth line contains N integers $C_1, C_2, \ldots, C_N$ ($0 \le C_i < 998244353$).

Output

For each $k = 1, 2, \ldots, N$, print the answer.

Example

standard input	standard output
3 1 2 3 4 5 6 7 8 9	146 1050 8694

Problem E. Edge Subsets

Input file:	*standard input*
Output file:	*standard output*
Time limit:	6 seconds
Memory limit:	1024 mebibytes

You are given integers A, B, and a simple undirected graph of N vertices and M edges. The vertices are numbered from 1 through N, and the edges from 1 through M. The edge i connects the vertices U_i and V_i. Here, it is guaranteed that $V_i - U_i = A$ or $V_i - U_i = B$.

Find the number of matchings of the graph, modulo 998244353. Note that a matching of the graph is a subset of edges whose end-points are all distinct.

Input

The first line contains integers N ($3 \le N \le 200$), M ($1 \le M \le 400$), A, and B ($1 \le A < B \le N - 1$).

The following M lines describe the edges. The i-th of those lines contains integers U_i and V_i ($1 \le U_i < V_i \le N$, $V_i - U_i = A$ or $V_i - U_i = B$). There are no self-loops or multi-edges.

Output

Print the answer.

Examples

standard input	standard output
4 3 1 2 1 2 1 3 3 4	5
10 14 2 4 5 7 7 9 2 6 6 8 1 5 3 7 4 8 1 3 4 6 8 10 3 5 5 9 2 4 6 10	225

Problem F. Find the LCA

Input file:	*standard input*
Output file:	*standard output*
Time limit:	7 seconds
Memory limit:	1024 mebibytes

You are given an integer sequence $A_1, A_2, \ldots, A_N$. You'll make a rooted tree with N vertices numbered from 1 through N. The vertex 1 is the root, and for each vertex i ($2 \le i \le N$), its parent p_i must satisfy $p_i < i$.

You define the score of a rooted tree as follows:

- Let x be the lowest common ancestor of the vertex $N-1$ and the vertex N. Then, the score is

$$\prod_{v \in (\text{subtree rooted at } x)} A_v$$

  Note that we consider x itself is in the subtree rooted at x.

There are $(N-1)!$ ways to make a tree. Find the sum of scores of all possible trees, modulo 998244353.

Input

The first line contains an integer N ($3 \le N \le 250000$).

The second line contains integers $A_1, A_2, \ldots, A_N$ ($1 \le A_i < 998244353$).

Output

Print the answer.

Examples

standard input	standard output
3 2 2 2	12
5 1 2 3 4 5	2080

Problem G. Games

Input file:	*standard input*
Output file:	*standard output*
Time limit:	4 seconds
Memory limit:	1024 mebibytes

You are given an integer sequence $A_1, A_2, \ldots, A_N$ and an integer K.

You'll prepare K piles of stones. Each pile should contain exactly A_i piles for some i. All piles are distinguishable; there are N^K different configurations.

You and Mike will play a game with the piles. You and Mike alternately do the following operation, with you going first.

- Choose at most 6 piles (choosing 0 piles is not allowed) and remove an arbitrary positive number of stones from each of the chosen piles. Note that the player can remove different numbers of stones from different piles.

The player who cannot make a valid move loses. Assuming both players play optimally, count the number of initial configurations that result in your loss, modulo 998244353.

Input

The first line contains integers N ($1 \le N \le 100$) and K ($1 \le K \le 10^{18}$).

The second line contains integers $A_1, A_2, \ldots, A_N$ ($1 \le A_1 < A_2 < \cdots < A_N \le 100$).

Output

Print the answer.

Examples

standard input	standard output
1 7 1	1
5 100 2 3 5 7 9	842434993

Problem H. Harsh Comments

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	1024 mebibytes

Posted on a blog are $N + M$ harsh comments. You made N comments, and the i-th of them has A_i downvotes. The i-th of the other M comments has B_i downvotes.

Mike is going to delete the comments, one by one, by repeating the following operation:

- Choose a comment randomly and delete it. More precisely, let $x_1, x_2, \ldots, x_k$ be numbers of downvotes the remaining comments have. Then, he will choose the i-th of them with the probability $x_i / \left(\sum_{1 \le j \le k} x_j\right)$ and detele it.

Note that the choices in the operations are independent.

Find the expected number of operations Mike will do until he deletes all of your comments. The answer is a rational number, so print it modulo 998244353 as usual. We can prove that such representation is always possible under the constraints of this problem.

Input

The first line contains integers N and M ($1 \le N, M \le 100$).

The second line contains integers $A_1, A_2, \ldots, A_N$ ($1 \le A_i \le 100$).

The third line contains integers $B_1, B_2, \ldots, B_M$ ($1 \le B_i$, $\sum_{1 \le i \le N} A_i + \sum_{1 \le i \le M} B_i < 998244353$).

Output

Print the answer.

Examples

standard input	standard output
1 2 1 1 1	2
1 1 2 1	332748119
3 3 2 3 5 7 11 900000000	636512475

Problem I. Inverse Problem

Input file:	*standard input*
Output file:	*standard output*
Time limit:	1 second
Memory limit:	1024 mebibytes

You are given an integer N and an integer sequence X of length M. Count, modulo 998244353, the number of permutations $P = (P_1, P_2, \ldots, P_N)$ of $(1, 2, \ldots, N)$ that satisfy the following condition:

- The lexicographically smallest subsequence of P of length M coincides with X.

Input

The first line contains integers N ($1 \le N \le 250000$) and M ($1 \le M \le N$).

The second line contains integers $X_1, X_2, \ldots, X_M$ ($1 \le X_i \le N$, $X_i \ne X_j$ for all $i \ne j$).

Output

Print the answer.

Examples

standard input	standard output
3 2 1 2	3
10 5 2 7 8 3 6	0
5 5 1 2 3 4 5	1

Problem J. Japanese Knowledge

Input file:	*standard input*
Output file:	*standard output*
Time limit:	10 seconds
Memory limit:	1024 mebibytes

This problem might be well-known in some countries, but how do other countries learn about such problems if nobody poses them?

You are given a **non-decreasing positive** integer sequence $A = (A_1, A_2, \ldots, A_N)$ of length N. For each $k = 0, 1, 2, \ldots, N$, count the number of **non-decreasing non-negative** integer sequences $x = (x_1, x_2, \ldots, x_N)$ of length N that satisfy following conditions, modulo 998244353:

- $x_i \le A_i$ for all $1 \le i \le N$.
- The number of indices i with $x_i = A_i$ is exactly k.

Input

The first line contains an integer N ($1 \le N \le 250000$).

The second line contains N integers $A_1, A_2, \ldots, A_N$ ($1 \le A_1 \le A_2 \le \cdots \le A_N \le 250000$).

Output

For each $k = 0, 1, 2, \ldots, N$, print the answer.

Examples

standard input	standard output
3 1 2 3	5 5 3 1
4 3 3 3 3	15 10 6 3 1
5 10 10 11 11 12	3982 1285 352 77 12 1